Maintenance Schedule

Diesel Engine
EPA/CARB
12V/16V4000S83
12V4000S83L
Application Group 4D

MSE50205/01E

A Rolls-Royce solution

Engine model	kW/cyl.	Application group
12V4000S83	139,8 kW/cyl.	4D, Frac-operation (continuous operation with low load)
12V4000S83L	155,4 kW/cyl.	4D, Frac-operation (continuous operation with low load)
16V4000S83	139,8 kW/cyl.	4D, Frac-operation (continuous operation with low load)

Table 1: Applicability

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DCL-ID: 0000042938 - 003

1 Maintenance Schedule

1.1 Preface

This manual provides instructions for your product from the manufacturer MTU.

Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section §15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

Maintenance concept

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: http://www.mtu-solutions.com.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emissions regulations. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations.

Although the warranty obligations are not dependent upon the use of any particular brand of replacement parts, it is recommended that emission-relevant components be serviced, replaced or repaired using only MTU-approved components or their equivalent. Emission-relevant components include, but are not limited to, control units, data records, sensors, injectors, exhaust turbo chargers, exhaust flaps and all components of the exhaust aftertreatment systems. Use of replacement parts that are not of equivalent quality to MTU-approved components may impair the effectiveness of emission control systems. The maintenance, replacement, or repair of the emission control devices and systems may be performed by any repair establishment or individual of the owner's choosing.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience.

Adherence to the specified maintenance intervals is essential to maintain product safety. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of MTU. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

- QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
- QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).
- QL3: Maintenance work which requires partial disassembly of the product.
- QL4: Maintenance work which requires complete disassembly of the product.

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Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

Interval (hrs)/limit	Limit in %	Limit max.
Operating hours	± 10%	± 500 hrs
Years	± 10%	± 30 days
Example 1	Task every 1000 hrs	Between 900 hrs and 1100 hrs (10% of 1000 hrs)
Example 2	Task every 30000 hrs	Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs)

Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the engine is to remain out of service for a longer time period, MTU recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at http://www.mtu-solutions.com.

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1.2 Allocation matrix application group 4D

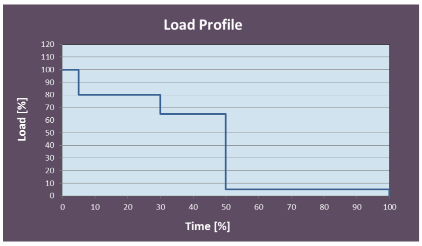
<table>
<tr><th colspan="2">Load profile</th><th colspan="7">V 4000 S83 / S83 L</th></tr>
<tr><th>Load %</th><th>Time %</th><th>Load factor %</th><th>Load indicator %</th><th>0 - 140 kW/cyl.</th><th>141 - 156 kW/cyl.</th><th></th><th></th><th></th></tr>
<tr><td>110 (100-110)</td><td>–</td><td rowspan="8">0 - 41</td><td rowspan="8">≤ 9</td><td rowspan="8">13,500</td><td rowspan="8">10,000</td><td rowspan="8">–</td><td rowspan="8">–</td><td rowspan="8">
</td></tr>
<tr><td>100 (90-100)</td><td>5</td></tr>
<tr><td>90 (80-90)</td><td>–</td></tr>
<tr><td>80 (65-80)</td><td>25</td></tr>
<tr><td>65 (50-65)</td><td>20</td></tr>
<tr><td>50 (30-50)</td><td>–</td></tr>
<tr><td>30 (5-30)</td><td>–</td></tr>
<tr><td>Idle (0-5)</td><td>50</td></tr>
</table>

Table 2: Allocation matrix application group 4D

TIM-ID: 0000061001 - 001

1.3 Maintenance schedule matrix 13,500 h

0 to 13,500 Operating hours

Task		Operating hours [h]																											
	Limit	Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500
Engine																													
W0800	-																												
W1024	1 w																												
W1008	2 a																												
W0500	-	X																											
W0501	-	X																											
W0503	-	X																											
W0505	-	X																											
W0506	-	X																											
W0525	-	X																											
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	6 m			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1463	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a							X						X						X						X			
W1207	2 a			X				X						X						X						X			
W1046	2 a										X									X									X
W1011	6 a										X									X									X
W1026	6 a										X									X									X
W1006	9 a										X									X									X
W1713	9 a										X									X									X
W1689	9 a										X									X									X
W1610	9 a																												X
W1250	5 a										X									X									X
W1251	6 a																												X
W2000	18 a																												X
W2001	18 a																												X
W2002	18 a																												X
W2004	18 a																												X
W2007	18 a																												X
W2009	18 a																												X
W2010	18 a																												X
W2062	18 a																												X
W2070	18 a																												X
W2110	18 a																												X
W2154	18 a																												X
W1051	18 a																												X
W1043	18 a																												X
W1134	18 a																												X
W1605	18 a																												X
W1735	18 a																												X
W1041	18 a																												X

w = weeks
m = months
a = years

TIM-ID: 0000070232 - 001

Task	Limit	Operating hours [h]																													
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500		
W1035	6 a																												X		
W3000	18 a																												X		
W3001	18 a																												X		
W3002	18 a																												X		
W3003	18 a																												X		
W3004	18 a																												X		
W3005	18 a																												X		
W3007	18 a																												X		
W3018	18 a																												X		
W3021	18 a																												X		
W3083	18 a																												X		
W3097	18 a																												X		
W3100	18 a																												X		
W3103	18 a																												X		
W3108	18 a																												X		
W3112	18 a																												X		
W3143	18 a																												X		
W3177	18 a																												X		
W3182	18 a																												X		
W1124	18 a																												X		
W1331	18 a																												X		
X5374	18 a																												X		

w = weeks
m = months
a = years

TIM-ID: 0000070232 - 001

1.4 Maintenance tasks 13,500 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).		W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Inspect battery-charging generator for contamination and clean if necessary.		W0525
QL1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.	X	W1009
QL1	1000	6 m	Automatic engine oil filter	Check and clean oil indicator filter.	X	W1864
QL1	1000	1 a	Engine mounts	Carry out visual inspection of engine mounts for general condition.	X	W1463
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.		W1241
QL1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL1	4500	2 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL1	4500	6 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL1	4500	6 a	Engine mounts	Check buffer clearance of mounts, check proper seating of securing screws.	X	W1026
QL1	4500	9 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	4500	9 a	Engine governor	Injector: reset drift compensation parameters (CDC).		W1713
QL1	4500	9 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL1	13500	9 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL3	4500	5 a	Rubber sleeves	Replace all rubber sleeves.		W1250
QL3	13500	6 a	Hose lines	Replace all hose lines.		W1251
QL3	13500	18 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.		W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Clean air ducting.		W2002
				Fit new high-pressure fuel sensor.		W2004
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.	X	W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110

w = weeks
m = months
a = years

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Qualifi-cation level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
				Check engine coolant thermostat and check thermal actuator, fit new one if necessary.		W2154
QL3	13500	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	13500	18 a	Battery charging alternator	Overhaul battery-charging generator.		W1043
QL3	13500	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	13500	18 a	Oil priming pump	Visually check oil priming pump.		W1605
QL3	13500	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	13500	18 a	Turbochargers	Fit new turbochargers.		W1041
QL4	13500	6 a	Automatic engine oil filter	Remove filter housing and clean. Fit new filter candles.	X	W1035
QL4	13500	18 a	Extended component mainte-nance	Completely disassemble the engine.Inspect engine components as per assembly instructions and repair or fit new components as re-quired.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace wiring harnesses.		W3083
				Fit new shaft misalignment coupling.	X	W3097
				Fit new rubber elements of engine mounts.	X	W3100
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Replace diaphragm coupling.	X	W3112
				Replace pulley.		W3143
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	13500	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	13500	18 a	Exhaust pipe bellows	Replace exhaust pipe bellows.		W1331
QL4	13500	18 a	Steel-spring coupling (Geisling-er)	Inspect steel-spring coupling, repair if necessary.	X	X5374
w = weeks m = months a = years						

TIM-ID: 0000070233 - 001

1.5 Maintenance schedule matrix 10,000 h

0 to 10,000 Operating hours

Task		Operating hours [h]																				
	Limit	Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000
Engine																						
W0800	-																					
W1024	1 w																					
W1008	2 a																					
W0500	-	X																				
W0501	-	X																				
W0503	-	X																				
W0505	-	X																				
W0506	-	X																				
W0525	-	X																				
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	6 m			X		X		X		X		X		X		X		X		X		X
W1463	1 a			X		X		X		X		X		X		X		X		X		X
W1001	2 a			X		X		X		X		X		X		X		X		X		X
W1241	2 a			X		X		X		X		X		X		X		X		X		X
W1575	1 a						X					X					X					X
W1207	2 a			X			X					X					X					X
W1046	2 a											X										X
W1011	6 a											X										X
W1026	6 a											X										X
W1006	9 a											X										X
W1713	9 a											X										X
W1689	9 a											X										X
W1610	9 a																					X
W1250	5 a											X										X
W1041	18 a											X										X
W1251	6 a																					X
W2000	18 a																					X
W2001	18 a																					X
W2002	18 a																					X
W2004	18 a																					X
W2007	18 a																					X
W2009	18 a																					X
W2010	18 a																					X
W2062	18 a																					X
W2070	18 a																					X
W2110	18 a																					X
W2154	18 a																					X
W1051	18 a																					X
W1043	18 a																					X
W1134	18 a																					X
W1605	18 a																					X
W1735	18 a																					X

w = weeks
m = months
a = years

TIM-ID: 0000070234 - 001

Task	Limit	Operating hours [h]																				
		Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000
W1035	6 a																					X
W3000	18 a																					X
W3001	18 a																					X
W3002	18 a																					X
W3003	18 a																					X
W3004	18 a																					X
W3005	18 a																					X
W3007	18 a																					X
W3018	18 a																					X
W3021	18 a																					X
W3083	18 a																					X
W3097	18 a																					X
W3100	18 a																					X
W3103	18 a																					X
W3108	18 a																					X
W3112	18 a																					X
W3143	18 a																					X
W3177	18 a																					X
W3182	18 a																					X
W1124	18 a																					X
W1331	18 a																					X
X5374	18 a																					X

w = weeks
m = months
a = years

TIM-ID: 0000070234 - 001

1.6 Maintenance tasks 10,000 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).		W1024
QL1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Inspect battery-charging generator for contamination and clean if necessary.		W0525
QL1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.	X	W1009
QL1	1000	6 m	Automatic engine oil filter	Check and clean oil indicator filter.	X	W1864
QL1	1000	1 a	Engine mounts	Carry out visual inspection of engine mounts for general condition.	X	W1463
QL1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.		W1241
QL1	2500	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL1	2500	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL1	5000	2 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL1	5000	6 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL1	5000	6 a	Engine mounts	Check buffer clearance of mounts, check proper seating of securing screws.	X	W1026
QL1	5000	9 a	Fuel injectors	Replace fuel injectors.		W1006
QL1	5000	9 a	Engine governor	Injector: reset drift compensation parameters (CDC).		W1713
QL1	5000	9 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL1	10000	9 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL3	5000	5 a	Rubber sleeves	Replace all rubber sleeves.		W1250
QL3	5000	18 a	Turbochargers	Fit new turbochargers.		W1041
QL3	10000	6 a	Hose lines	Replace all hose lines.		W1251
QL3	10000	18 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.		W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Clean air ducting.		W2002
				Fit new high-pressure fuel sensor.		W2004
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.	X	W2009
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110

w = weeks
m = months
a = years

TIM-ID: 0000070235 - 001

Qualifi-cation level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
				Check engine coolant thermostat and check thermal actuator, fit new one if necessary.		W2154
QL3	10000	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	10000	18 a	Battery charging alternator	Overhaul battery-charging generator.		W1043
QL3	10000	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	10000	18 a	Oil priming pump	Visually check oil priming pump.		W1605
QL3	10000	18 a	Auxiliary PTO	Replace PTO.		W1735
QL4	10000	6 a	Automatic engine oil filter	Remove filter housing and clean. Fit new filter candles.	X	W1035
QL4	10000	18 a	Extended component maintenance	Completely disassemble the engine.Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace wiring harnesses.		W3083
				Fit new shaft misalignment coupling.	X	W3097
				Fit new rubber elements of engine mounts.	X	W3100
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Replace diaphragm coupling.	X	W3112
				Replace pulley.		W3143
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	10000	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	10000	18 a	Exhaust pipe bellows	Replace exhaust pipe bellows.		W1331
QL4	10000	18 a	Steel-spring coupling (Geislinger)	Inspect steel-spring coupling, repair if necessary.	X	X5374

w = weeks
m = months
a = years

TIM-ID: 0000070235 - 001